

Chapter 11: Azure AI Foundry

Learning Objectives

By the end of this chapter, you will:

- Understand the capabilities of Azure AI Foundry and its role in the enterprise AI ecosystem.
- Learn how Azure AI Foundry supports model selection, prompt management, evaluations, and AI Agent development.
- Understand how AI TPMs can use Azure AI Foundry to accelerate AI delivery and governance.

Beginner Explanation

Azure AI Foundry is Microsoft's unified platform for building, testing, evaluating, and managing AI applications.

Think of Azure AI Foundry as an "AI Control Center."

Instead of managing multiple tools separately, Azure AI Foundry provides a single place to:

- Explore AI models
- Build AI applications
- Create AI agents
- Manage prompts
- Run evaluations
- Monitor performance
- Govern AI systems

Simple Analogy

Azure AI Foundry Component	Analogy
Model Catalog	App Store
Prompt Flow	Workflow Designer
Evaluations	Quality Inspector
Agents	Digital Employees
Monitoring	Control Room
Projects	Team Workspace

What Can You Build?

- Knowledge Assistants
- Customer Support Bots
- AI Agents

- RAG Applications
- Document Processing Systems
- Executive Dashboards

For an AI TPM, Azure AI Foundry simplifies AI program delivery by centralizing AI development activities.

Intermediate Explanation

Azure AI Foundry enables organizations to move from experimentation to production more efficiently.

Core Capabilities

Capability	Description
Model Catalog	Access to Microsoft and third-party models
Prompt Flow	Build and test workflows
Evaluations	Measure AI quality and performance
Agents	Create autonomous AI systems
Monitoring	Track usage and performance
Governance	Manage enterprise AI policies

Enterprise Delivery Workflow

1. Create a project.
2. Select a model.
3. Build prompt flows.
4. Configure evaluations.
5. Test with users.
6. Deploy to production.
7. Monitor continuously.

Common Enterprise Use Cases

Use Case	Example
RAG	HR policy assistant
AI Agents	Incident management
Evaluations	Hallucination tracking
Prompt Management	Executive reports
Multi-Model Testing	Cost optimization

Benefits

- Faster delivery
- Improved governance

- Better visibility
- Centralized management
- Reduced operational complexity

Why This Matters for an AI TPM

Responsibility	Impact
Delivery	Accelerates AI development and deployment across teams.
Risk	Provides evaluation and governance capabilities.
Stakeholder Management	Enables transparency through dashboards and reporting.
Budget	Helps optimize model selection and operational costs.

Enterprise Example

Business Problem

A healthcare provider wanted to build multiple AI solutions, including:

- Patient support assistant
- Internal knowledge assistant
- Claims processing agent
- Executive reporting dashboard

Managing these projects independently created operational challenges.

Solution

Adopt Azure AI Foundry as the centralized AI platform.

Architecture

- Azure AI Foundry
- Azure OpenAI
- Azure AI Search
- Azure Functions
- Azure Monitor
- Power BI

Outcome

Metric	Result
Time to Production	Reduced by 35%
AI Projects Managed	12

Metric	Result
User Satisfaction	93%
Governance Compliance	100%

Lessons Learned

- Centralized governance improves consistency.
- Evaluations significantly reduce production issues.
- Shared prompt libraries improve productivity.
- Multi-project visibility helps leadership teams.

Azure Implementation

Services Used

Service	Purpose
Azure AI Foundry	Central AI development platform
Azure OpenAI	Model inference
Azure AI Search	Knowledge retrieval
Azure Functions	Workflow orchestration

Implementation Workflow

Step 1: Create an AI Foundry Project

Define:

- Project name
- Owners
- Stakeholders
- Success metrics

Step 2: Select Models

Examples:

Model Type	Use Case
GPT	Text generation
Embedding Models	RAG
Vision Models	Image analysis

Model Type	Use Case
Speech Models	Voice applications

Step 3: Build Prompt Flows

Create workflows for:

- Summarization
- Document analysis
- Risk assessments
- Report generation

Step 4: Configure Evaluations

Track:

- Accuracy
- Latency
- Hallucinations
- Cost
- User satisfaction

Step 5: Deploy and Monitor

Monitor:

- Usage
- Performance
- Errors
- Monthly costs
- Adoption rates

Mermaid Diagram

```
```mermaid id="1i8hmg" graph TD
```

```
A[Business Requirements]
```

```
A --> B[Azure AI Foundry]
```

```
B --> C[Model Catalog]
```

```
B --> D[Prompt Flows]
```

B --> E[Evaluations]

B --> F[AI Agents]

C --> G[Azure OpenAI]

D --> G

E --> G

F --> G

G --> H[Enterprise Applications]

```

Azure AI Foundry Delivery Workflow

```mermaid id="w0ztyn"
graph LR
  A[Ideation]
  A --> B[Project Creation]
  B --> C[Model Selection]
  C --> D[Prompt Development]
  D --> E[Evaluations]
  E --> F[Production Deployment]
  F --> G[Monitoring]
```

TPM Checklist

- ☐ Create AI Foundry projects.
- ☐ Define business objectives.
- ☐ Select appropriate models.
- ☐ Configure prompt flows.
- ☐ Establish evaluation criteria.
- ☐ Enable monitoring.
- ☐ Estimate operational costs.
- ☐ Implement governance controls.
- ☐ Review KPIs monthly.
- ☐ Obtain stakeholder approval.

Templates

AI Foundry Project Template

Field	Description
Project Name	
Business Objective	
Sponsor	
TPM	
KPIs	
Budget	

Evaluation Dashboard

Metric	Target
Accuracy	>95%
Latency	<3 Seconds
User Satisfaction	>90%
Availability	99.9%
Cost	Within Budget

Risk Register

Risk	Impact	Mitigation
Model Selection Issues	Medium	Evaluation framework
Governance Gaps	High	Approval workflows
Cost Overruns	Medium	Budget monitoring
Low Adoption	Medium	User enablement

Common Mistakes

1. Treating Azure AI Foundry as only a development tool.
2. Skipping evaluations before production deployment.
3. Ignoring governance capabilities.
4. Failing to standardize prompt flows.

5. Underestimating operational costs.
 6. Neglecting monitoring and reporting.
 7. Managing AI projects independently instead of centrally.
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Key Takeaways

- Azure AI Foundry is Microsoft's centralized platform for enterprise AI development.
 - It provides capabilities for model management, prompt engineering, evaluations, and AI Agents.
 - Centralized governance improves consistency and reduces operational risks.
 - AI TPMs can use Azure AI Foundry to improve delivery velocity and visibility.
 - Evaluations and monitoring are essential for successful production deployments.
 - Azure AI Foundry should be considered a strategic platform for organizations adopting AI at scale.
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Footer
